

Sam Gunn

CONTACT

Email: gunn@mit.edu

Website: <https://samuel-gunn.github.io/>

POSITIONS

Massachusetts Institute of Technology

C.L.E. Moore Instructor of Mathematics (Summer 2026 to Present)

EDUCATION

The University of California, Berkeley

Ph.D. in Computer Science (Fall 2020 to Summer 2026)

Advisor: Umesh Vazirani

The University of Texas at Austin

B.S. in Mathematics (Fall 2016 to Spring 2020)

Advisors: Scott Aaronson and Cameron Gordon

SELECTED WORKS

1. **How to sketch a learning algorithm.** *In submission.* [arXiv](#).
2. **Improved pseudorandom codes from permuted puzzles.** With Miranda Christ, Noah Golowich, Ankur Moitra, and Daniel Wichs. *In submission.* [ePrint](#).
3. **Black-box crypto is useless for pseudorandom codes.** With Sanjam Garg and Mingyuan Wang. *TCC 2025 (outstanding paper award).* [ePrint](#).
4. **An undetectable watermark for generative image models.** With Xuandong Zhao and Dawn Song. *ICLR 2025.* [arXiv](#).
5. **SoK: Watermarking for AI-generated content.** With Xuandong Zhao et al. *S&P 2025.* [arXiv](#).
6. **Ideal pseudorandom codes.** With Omar Alrabiah, Prabhanjan Ananth, Miranda Christ, and Yevgeniy Dodis. *STOC 2025 (invited to the SICOMP Special Issue).* [ePrint](#).
7. **Quantum one-time protection of any randomized algorithm.** With Ramis Movassagh. *CRYPTO 2025.* [ePrint](#).
8. **Classical commitments to quantum states.** With Yael Kalai, Anand Natarajan, and Agi Villanyi. *QIP 2025, STOC 2025.* [ePrint](#).
9. **Pseudorandom error-correcting codes.** With Miranda Christ. *CRYPTO 2024.* [ePrint](#).
10. **Undetectable watermarks for language models.** With Miranda Christ and Or Zamir. *COLT 2024, RWC 2024.* [ePrint](#).
11. **How to use quantum indistinguishability obfuscation.** With Andrea Coladangelo. *STOC 2024.* [ePrint](#).
12. **Approaching the quantum Singleton bound with approximate error correction.** With Thiago Bergamaschi and Louis Golowich. *QIP 2024, STOC 2024.* [arXiv](#).
13. **Commitments to quantum states.** With Nathan Ju, Fermi Ma, and Mark Zhandry. *QIP 2023, STOC 2023 (invited to the SICOMP Special Issue).* [ePrint](#).

RECOGNITION

1. Google PhD Fellowship 2022–2025
2. Lotfi A. Zadeh Prize 2024–2025
3. Simons Institute annual magazine: [link](#)
4. Communications of the ACM: [link](#)
5. Outstanding Paper Award at TCC 2025
6. Invitations to the SICOMP Special Issues for STOC 2023 and 2025 (reserved for the top 10% of accepted STOC papers)

TALKS

1. **How to sketch a learning algorithm**
 - MIT Crypto+AI seminar (May 2026).
 - Institute for Advanced Study (May 2026).
 - Flatiron Institute (June 2026).
 - STOC Theory for Trustworthy and Interpretable AI Workshop (June 2026).
 - Invited talk at ITC (August 2026).
2. **On undetectable backdoors** at the UT Austin Theory and Alignment group (November 2025).
3. **Pseudorandom error-correcting codes**
 - NTT Research (March 2024).
 - Google Research NYC (April 2024).
 - CRYPTO 2024 (August 2024).
 - NYU (March 2025).
 - Harvard Theory Seminar (September 2025).
 - TCC 2025 (December 2025).
4. **Undetectable watermarks for language models**
 - Google Research Mountain View (July 2023).
 - MIT Cryptography Group (August 2023).
 - NYU Cryptography Seminar (September 2023). Video [here](#).
 - Berkeley Machine Learning Theory Seminar (September 2023).
5. **Quantum one-time protection of any randomized algorithm** at CRYPTO 2025 (August 2025).
6. **Quantum indistinguishability obfuscation**
 - CIQC Spark Talk, March 2024.
 - The University of Ottawa, April 2024.
7. **Approaching the quantum Singleton bound with approximate error correction** at the Simons Quantum Colloquium (December 2022). Video [here](#).
8. **Commitments to quantum states** at STOC 2023 (June 2023). Video [here](#).
9. **Unclonable cryptography** guest lecture for John Wright’s graduate quantum complexity course at UC Berkeley (December 2022).
10. **Certified entropy from random circuit sampling** at Berkeley Theory Lunch (March 2022).
11. **On the classical hardness of spoofing linear cross-entropy benchmarking** plenary talk at SQuInT 2020 (February 2020).